Properties of 2D Shapes

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 95 minutes · Total marks: 95

KEY RULES

Triangles	Equilateral, isosceles, scalene, right-angled
Quadrilaterals	Square, rectangle, rhombus, parallelogram, kite, trapezium
Symmetry	Lines of symmetry and rotational symmetry order
Polygons	Regular: equal sides and equal angles; interior $= \dfrac{(n-2)180^\circ}{n}$

WORKED EXAMPLE 1

[2 marks]

Work out the sum of the interior angles of a pentagon.

Step 1 Sum = $(n-2) \times 180$ $(5-2) \times 180 = 540$

Answer: 540°

WORKED EXAMPLE 2

[3 marks]

Each exterior angle of a regular polygon is 24°. How many sides does it have?

Step 1 Number of sides = $360 \div$ exterior angle $360 \div 24 = 15$

Answer: 15

WORKED EXAMPLE 3

[3 marks]

ABCD **is a rhombus. Angle** *ABC* $= 70^\circ$**. Work out angle** *BCD*.

Step 1 Adjacent angles in a rhombus add to 180 $180 - 70$

Answer: 110°

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Properties of 2D Shapes

1 Each exterior angle of a regular polygon is 24°

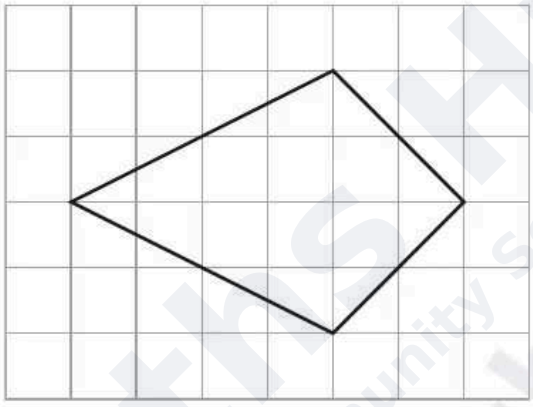
Work out the number of sides of this regular polygon.

.....

(Total for question = 2 marks)

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2 **10** The diagram shows a kite drawn on a centimetre grid.



On the centimetre grid below, draw a rectangle that has the same area as the kite.

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3 Each interior angle of a regular polygon is 156°

Work out the number of sides of the polygon.

.....

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4 *ABCDE* is a pentagon.

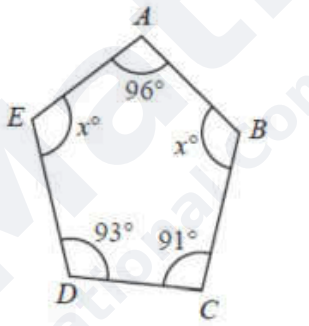


Diagram **NOT** accurately drawn

angle *EAB* = 96° angle *BCD* = 91° angle *CDE* = 93° angle *DEA* = angle *ABC* = *x*°

Work out the value of *x*

x = ...

(Total for question = 3 marks)

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5 Here is a 9-sided regular polygon $ABCDEFGHJ$, with centre O

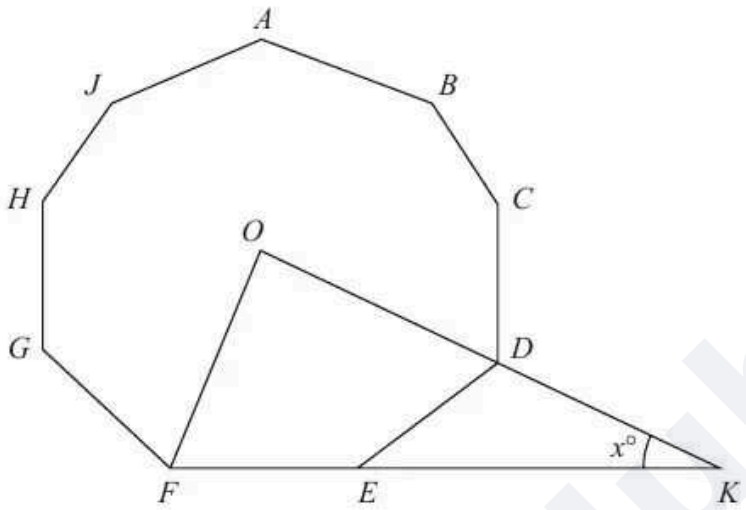


Diagram **NOT** accurately drawn

ODK and FEK are straight lines.

Work out the value of x

$x =$ .....

(Total for Question 11 is 3 marks)

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6 19



Diagram **NOT** accurately drawn

The diagram shows part of a regular polygon.
The interior angle and the exterior angle at a vertex are marked.
The size of the interior angle is 7 times the size of the exterior angle.

Work out the number of sides of the polygon.

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7 Here is a quadrilateral.

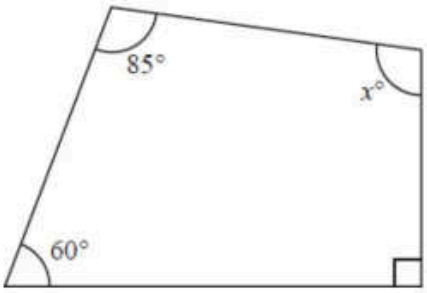


Diagram **NOT** accurately drawn

(a) Work out the value of x

$x =$ .....

(2)

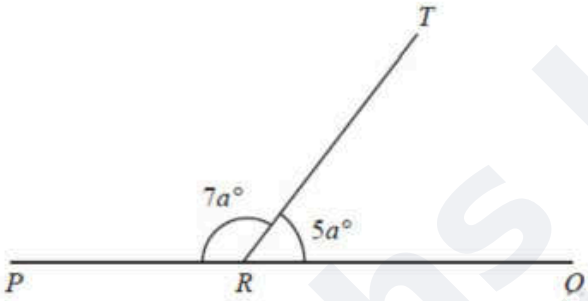


Diagram **NOT** accurately drawn

PRQ is a straight line.

(b) Work out the value of a

$a =$ .....

(2)

(Total for question = 4 marks)

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8 7

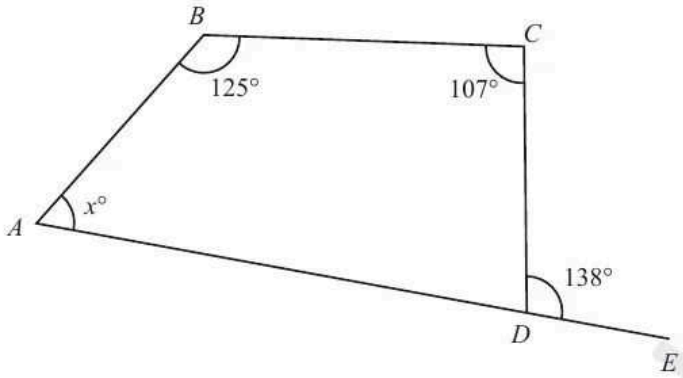


Diagram **NOT** accurately drawn

ABCD is a quadrilateral.

ADE is a straight line.

Work out the value of x
Give a reason for each stage of your working.

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9 9 Here is a quadrilateral.

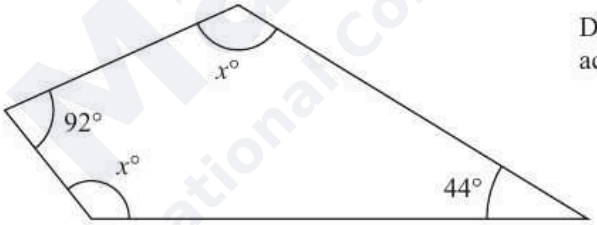


Diagram **NOT** accurately drawn

(i) Work out the value of x

(3)

(ii) Give a reason for your answer.

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10

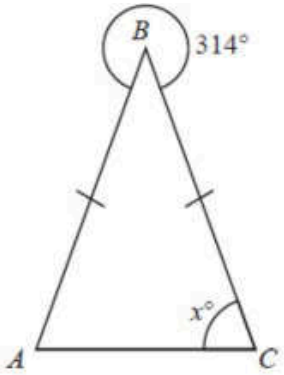


Diagram **NOT** accurately drawn

ABC is an isosceles triangle.

$AB = CB$ reflex angle $ABC = 314°$ angle$ACB = x°$

Work out the value of x
Give a reason for each stage of your working.

$x =$ .....

(Total for question = 4 marks)

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11 6

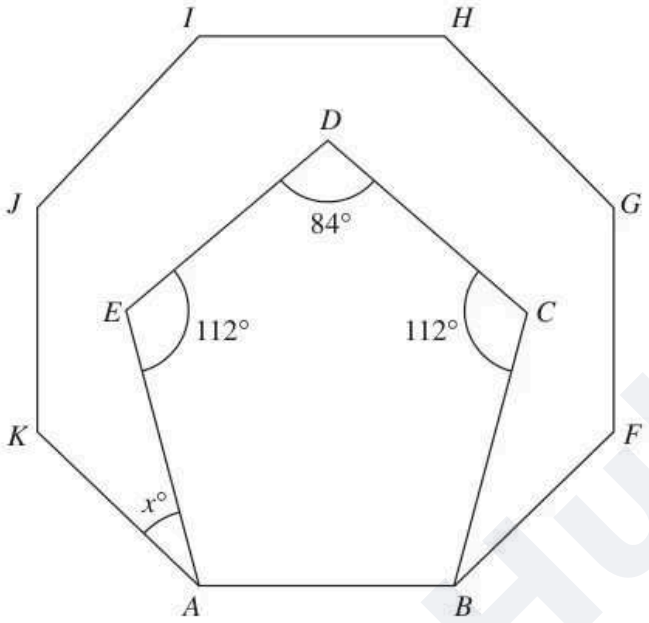


Diagram **NOT** accurately drawn

Pentagon *ABCDE* is drawn inside the regular octagon *ABFGHIJK*.
The pentagon has exactly one line of symmetry.

Work out the value of x.

$x =$

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12 9

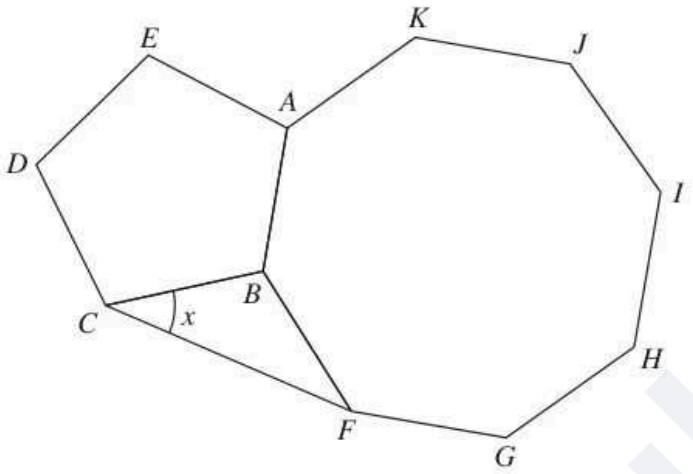


Diagram **NOT** accurately drawn

The diagram shows a regular pentagon, *ABCDE*, a regular octagon, *ABFGHIJK*, and an isosceles triangle, *BCF*.

Work out the size of angle x.

.....°

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13 (a) Find the sum of the interior angles of a polygon with 7 sides.

.....°

(2)

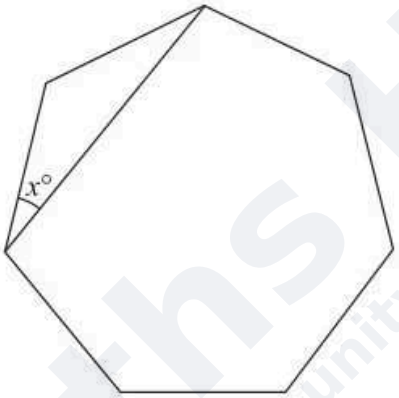


Diagram **NOT** accurately drawn

The diagram shows a regular polygon with 7 sides.

(b) Work out the value of x.
Give your answer correct to 1 decimal place.

.....

(2)

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14 The diagram shows a regular 10-sided polygon, *ABCDEFGHIJ*

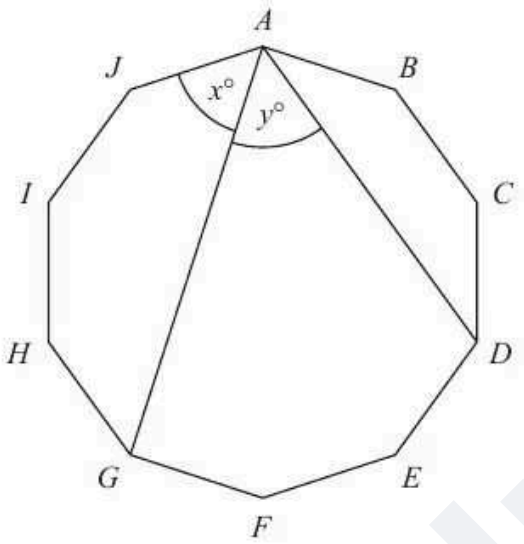


Diagram **NOT** accurately drawn

Show that $x = y$

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15 12

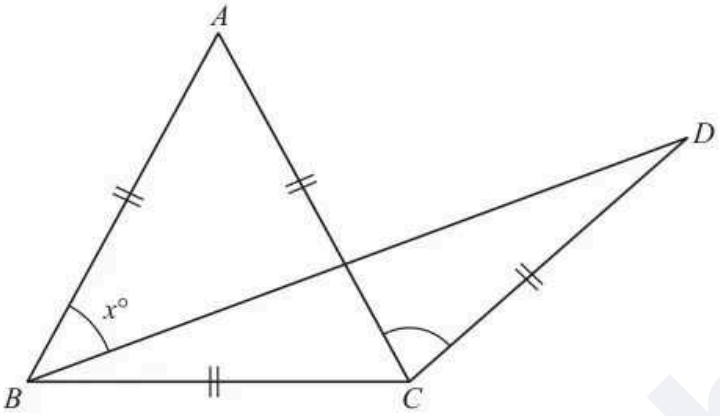


Diagram **NOT** accurately drawn

The diagram shows an equilateral triangle ABC and an isosceles triangle BCD.
$AB = AC = BC = CD$.
Angle $ABD = x°$

Express the size of angle ACD in terms of $x°$, giving your answer as simply as possible.
Give a reason for each step in your working.

.....°

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16 The diagram shows an incomplete regular polygon.

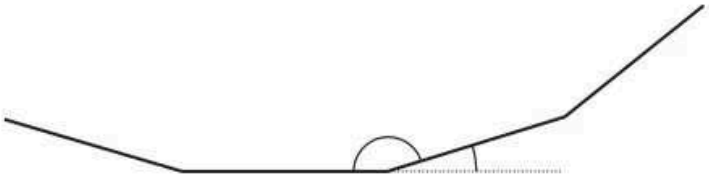


Diagram **NOT** accurately drawn

The size of each interior angle is 140 degrees greater than the size of each exterior angle.

Work out the number of sides the regular polygon has.

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17 Here is a regular 10-sided polygon.

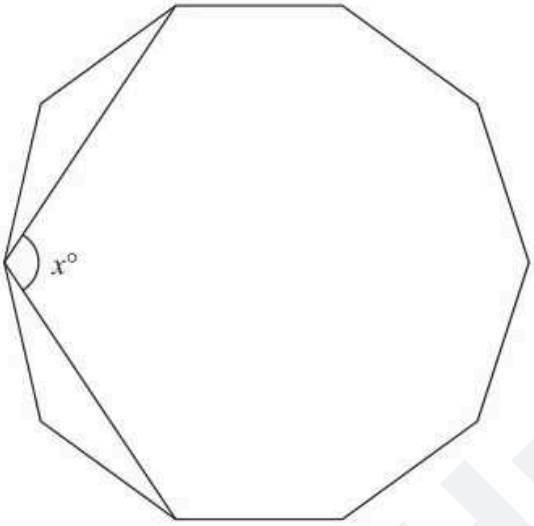


Diagram **NOT** accurately drawn

Work out the value of x.
Show your working clearly.

$x =$ .....

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18 15

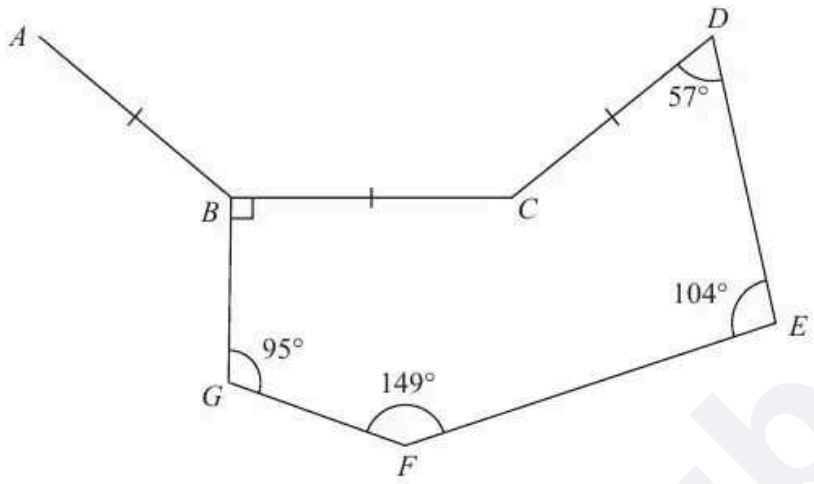


Diagram **NOT** accurately drawn

BCDEFG is a hexagon.
AB, *BC* and *CD* are three sides of a regular *n*-sided polygon.

Calculate the value of *n*
Show your working clearly.

n = ...

(Total for Question 15 is 4 marks)

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19 4

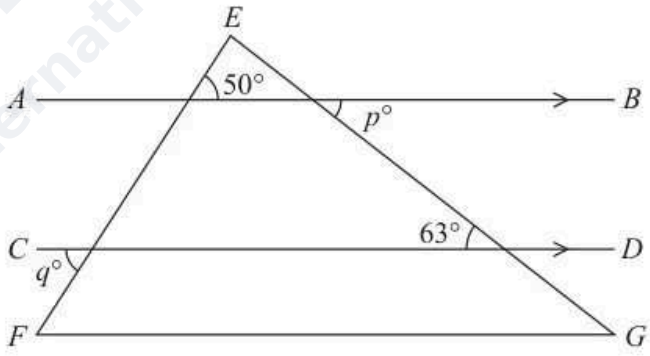


Diagram **NOT** accurately drawn

EFG is a triangle.
AB is parallel to *CD*.

(a) Write down the value of *p*

p = ...
(1)

(b) Write down the value of *q*

q = ...
(1)

Here is a hexagon.

CHALLENGE QUESTIONS

20 13

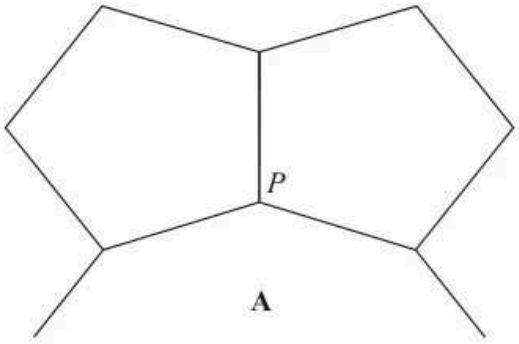


Diagram **NOT** accurately drawn

The diagram shows two congruent regular pentagons and part of a regular n-sided polygon **A**.
Two sides of each of the regular pentagons and two sides of **A** meet at the point P.

Calculate the value of n.
Show your working clearly.

$n =$ .....

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21 **12**

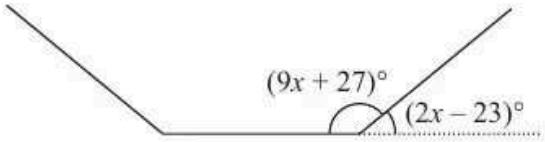


Diagram **NOT** accurately drawn

The diagram shows part of a regular n-sided polygon with

an interior angle of $(9x + 27)°$
an exterior angle of $(2x – 23)°$

Work out the value of n

$n =$ .....

(Total for Question 12 is 4 marks)

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22 **3**

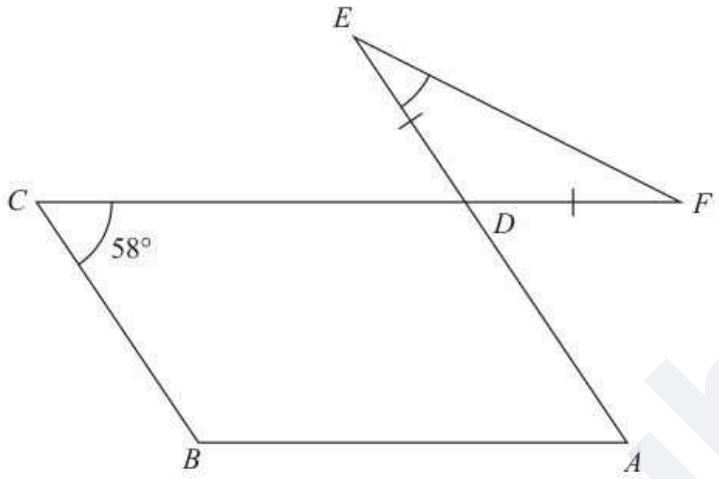


Diagram **NOT** accurately drawn

The diagram shows a parallelogram $ABCD$ and an isosceles triangle DEF in which $DE = DF$

CDF and ADE are straight lines.
Angle $BCD = 58°$

Work out the size of angle DEF.
Give a reason for each stage of your working.

.....°

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23 The diagram shows parallelogram *EFGH*.

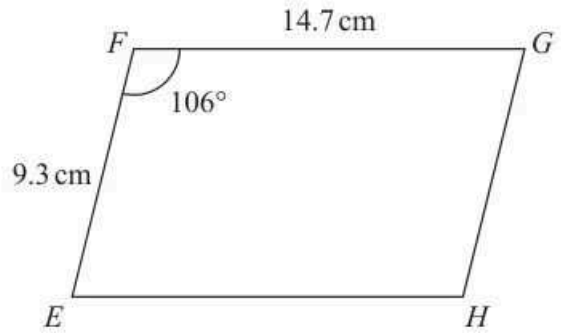


Diagram **NOT** accurately drawn

EF = 9.3 cm
FG = 14.7 cm
Angle *EFG* = 106°

(a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm²
(2)

(b) Work out the length of the diagonal *EG* of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

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24 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

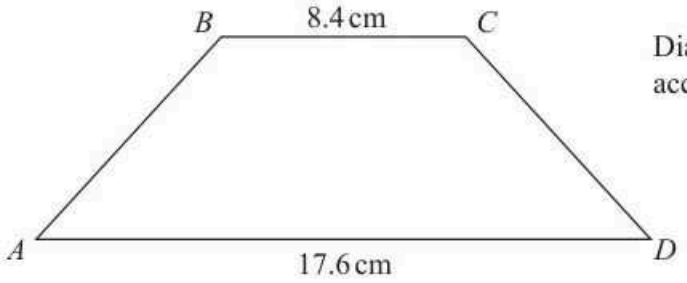


Diagram **NOT** accurately drawn

The trapezium has exactly one line of symmetry.

$BC = 8.4\,\text{cm}$
$AD = 17.6\,\text{cm}$

The trapezium has area $179.4\,\text{cm}^2$

Work out the size of angle ABC.
Give your answer correct to 1 decimal place.

.....°

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TOTAL FOR PAPER IS 95 MARKS

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